EXPERIMENT NO : 2

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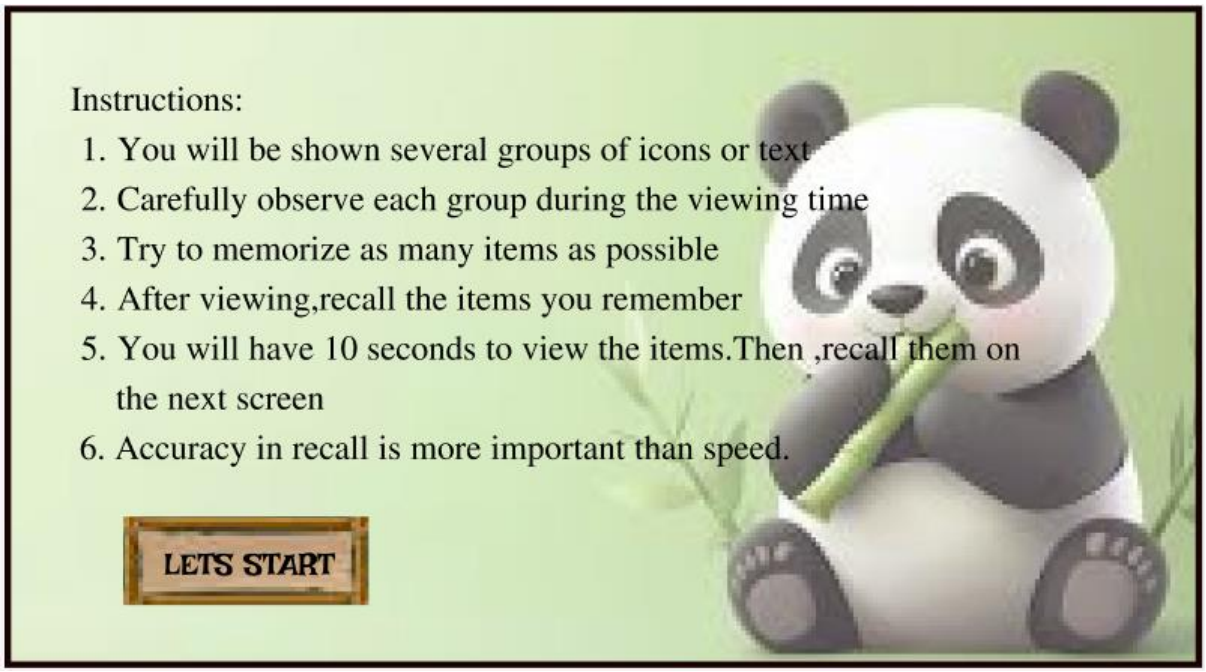
Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Screen

Chunking is a cognitive technique that breaks information into small, meaningful units to improve understanding and memory retention. The instruction screen in this design effectively applies chunking as follows:

1. **Clear Step-by-Step Numbering**
   - Instructions are divided into **six numbered steps**, each conveying one clear action.
   - This reduces cognitive overload and helps users follow the task easily.
2. **Logical Information Grouping**
   - **Viewing Phase** (Steps 1–2): Explains what the user will see.
   - **Memory Phase** (Step 3): Encourages memorization.
   - **Recall Phase** (Steps 4–5): Explains how recall will happen.
   - **Performance Rule** (Step 6): Emphasizes accuracy over speed.
3. **Visual Hierarchy and Design**
   - Large instruction heading improves visibility.
   - Proper spacing and alignment increase readability.
   - The **“LET’S START” button** clearly indicates the next action.
   - Panda illustration creates a friendly and engaging atmosphere.
4. **Time Expectation Reinforcement**
   - Users are informed that they will get **10 seconds** to view the items, setting clear expectations.
5. **Simplicity and Clarity**
   - Short, simple sentences ensure easy comprehension for all users.



FRAME 2: CHUNKING (MEMORIZATION) PHASE

Analysis of the Chunking Phase Screen

This screen represents the **visual encoding stage**, where users observe and memorize grouped icons.

1. **Purpose of the Screen**
    - Allows users to visually scan and mentally group panda-emoji icons within a limited time.
    - Encourages memory strategies such as pattern recognition and grouping.
2. **Key UI Elements**

⏳ **Countdown Timer**

- Displays remaining time (e.g., 7 seconds).
- Creates urgency and keeps users focused.

📊 **Progress Bar**

- Visually represents time passing.
- Reinforces the limited viewing duration.

▯ **Grid of Panda Emojis**

- Panda icons with different facial expressions are arranged in a structured grid.

- Similar expressions allow users to form memory chunks (happy, sad, angry, sleepy).

🐼 **Thematic Panda Background**

- Maintains visual consistency and reduces distraction.

3. **How Chunking Works Here**
    - Users group emojis based on facial expressions.
    - Chunking helps remember groups instead of individual icons.
4. **Cognitive Benefits**
    - Improves short-term memory.
    - Reduces mental load.
    - Enhances recall accuracy.

FRAME 3: RECALL (SELECTION) PHASE

Analysis of the Recall Phase Screen

This screen tests how effectively users can retrieve stored visual information.

1. **Purpose of the Screen**

    - Measures the user's memory by asking them to identify previously seen icons.

2. **Key UI Components**

▯ Recall Options Grid

- Displays panda emojis, including:
    - Correct icons from the chunking phase
    - Distractor icons not shown earlier

⭕ Selection Indicators

- Users select remembered icons using radio buttons or check marks.

🐼 Interactive Panda Characters

- Visual feedback (tick marks) confirms correct selections.
- Panda theme maintains engagement.

➡️ NEXT Button

- Moves the user to the result screen.

3. **How the Recall Phase Works**
    - Users compare memory with presented options.
    - Select only icons they remember.
    - Distractors test false memory recognition.
4. **UX and Cognitive Benefits**
    - Evaluates memory accuracy.
    - Prevents guessing through distractors.
    - Simple interaction reduces confusion.

FRAME 4: RESULT / SCORE PAGE

Analysis of the Score & Feedback Screen

This screen provides **performance feedback** after the recall task.

1. **Purpose of the Screen**
    - Displays the user's recall score.
    - Gives a sense of achievement and closure.
2. **Key UI Elements**

📋 **Score Display Panel**

- Shows "Your Score: 3" clearly and prominently.
- Wooden board design gives a game-like feel.

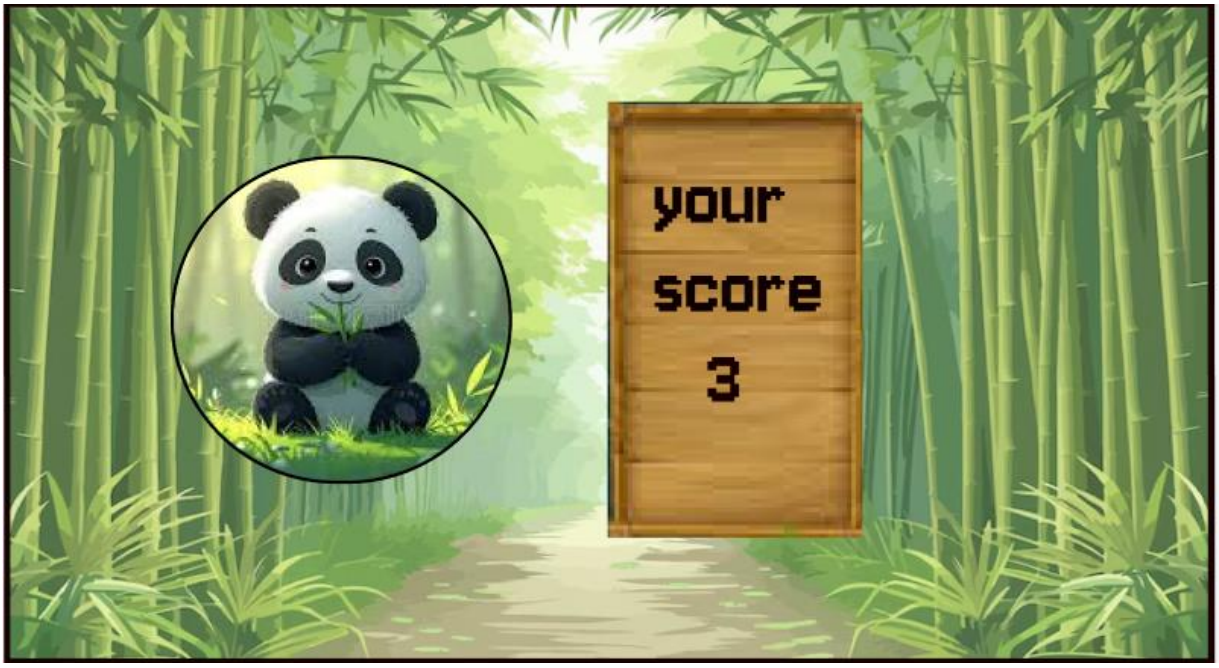
🐼 **Panda Avatar**

- Reinforces friendly feedback instead of pressure.

🌿 **Background Environment**

- Calm bamboo forest theme reduces stress.

3. **How Feedback Works**
    - System counts correct selections.

- Score reflects recall performance.

4. **UX Benefits**
    - Instant feedback improves learning.
    - Encourages replay and self-improvement.
    - Gamified visuals maintain motivation.